



Consumer Attention Mapping System

AI-powered retail attention
intelligence platform

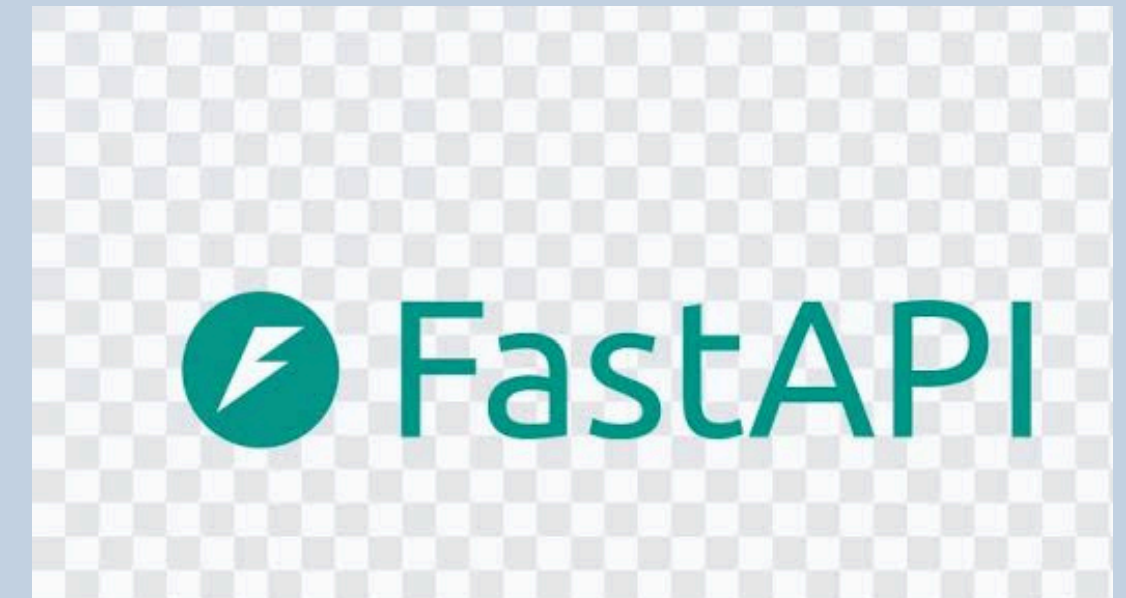
Technology Stack



Python is a high-level, interpreted programming language known for its simple and readable syntax. It is widely used for web development, automation, data analysis, artificial intelligence, and machine learning.



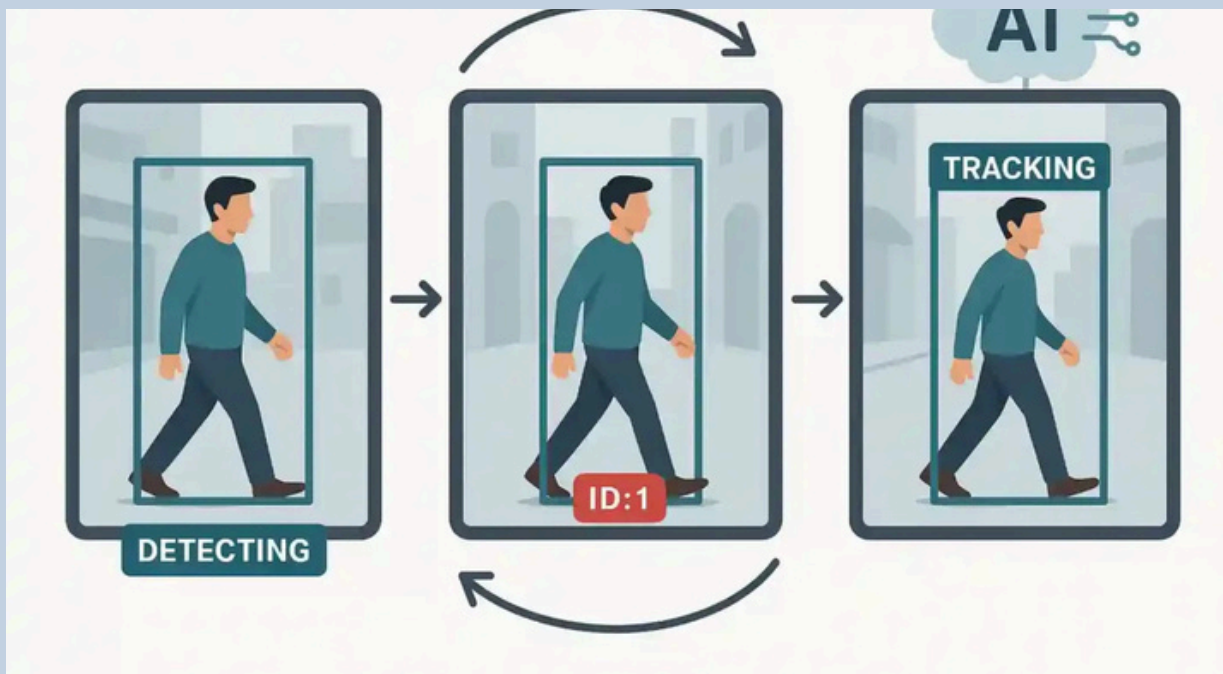
React.js is a JavaScript library used for building interactive and dynamic user interfaces. It helps developers create reusable UI components and efficiently build modern web applications.



FastAPI is a modern, high-performance Python framework used for building APIs and backend applications. It provides fast API development with automatic documentation, data validation,

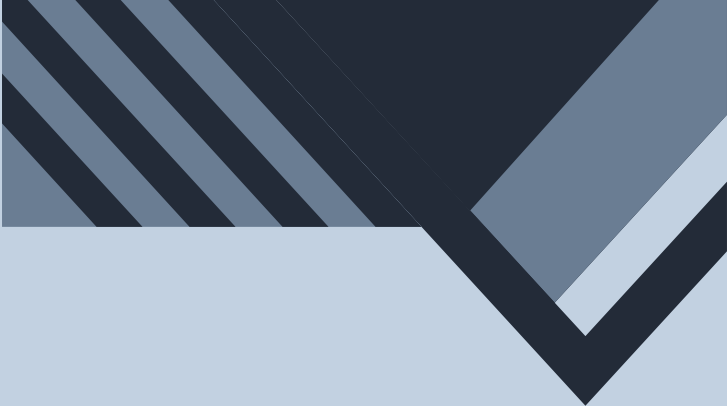
Core Features

Shopper Detection & Tracking – multi-person, path, zone analysis



Product Interaction – pickup, return, purchase detection





Challenges in Retail Analytics

Limited Insights:

retailers lack real-time attention data Most retailers rely on sales numbers or basic footfall counts, which fail to capture why customers engage with certain products. Without visibility into attention patterns, they miss opportunities to optimize shelf placement and promotions.

Manual Observation – inefficient, error-prone, low accuracy:

Traditional methods like staff observation or surveys are slow, subjective, and prone to human error. They cannot scale across multiple stores or provide consistent, actionable insights in real time.

Poor Shelf Placement :

reduces visibility and conversions Products often go unnoticed due to suboptimal shelf positioning. This leads to wasted marketing spend and lower conversion rates, as retailers cannot identify which placements truly drive customer engagement.

Solution:

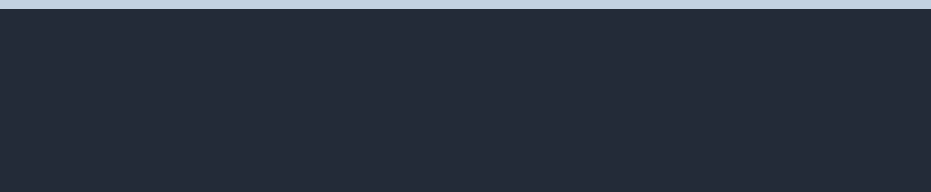
AI-powered platform for automated, real-time consumer analytics By leveraging cameras, computer vision, and AI models, the system delivers objective, scalable, and real-time insights. Retailers gain actionable intelligence to improve product visibility, optimize layouts, and boost conversions.

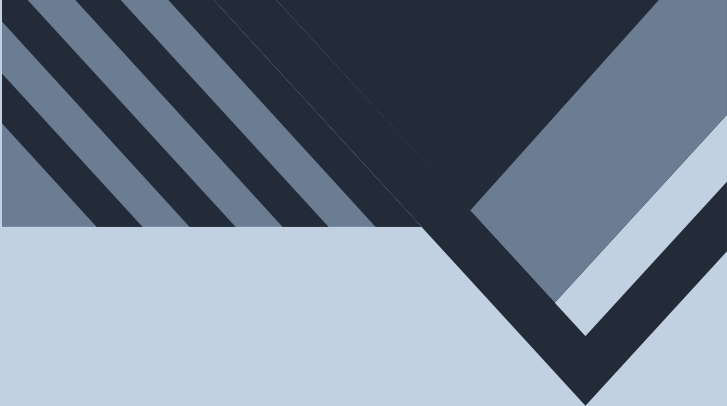




System Architecture

The architecture of the Consumer Attention Mapping System is designed for robustness, scalability, and security. At the foundation, data sources such as in-store cameras, POS systems, CRM platforms, and cloud storage feed information into the system. This data is processed through a microservices layer that handles shopper detection, tracking, attention analysis, and product interaction, ensuring each function operates independently for easy updates and scalability. The processed information flows into the analytics layer, where heatmaps, attractiveness scoring, and optimization recommendations are generated to provide actionable insights. Finally, these insights are delivered through intuitive dashboards tailored for different roles — Store Managers, Retail Analysts, Marketing Managers, and Administrators — enabling each stakeholder to monitor performance, understand consumer behavior, and make data-driven decisions in real time.





Challenges & Solutions

Scalability:

Modular microservices → seamless growth

Security:

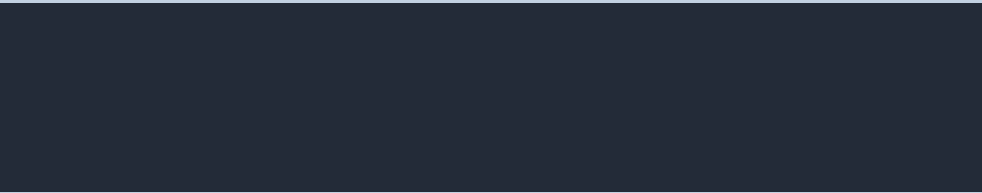
JWT, OAuth2, encryption → safe data handling

Engagement:

Real-time dashboards → actionable insights

Integration:

APIs for POS, CRM, marketing platforms → smooth interoperability



Frontend Outputs

Consumer Attention System

Welcome Back!

Email*

Enter your email

Password*

Enter your password

Login

Don't have an account?Register Here

Consumer Attention System

Create New Account

Username*

Enter Username

Email*

Enter Email

Mobile Number*

Enter Mobile Number

Password*

Enter Password

Confirm Password*

Confirm Password

Register

Already have an account?Login Here

Consumer Attention Mapping

Dashboard

Stores

Shelves

Cameras

Consumers

Reports

Settings

Logout

Welcome back, Store Manager!

Here's what's happening in your stores today.

Stores

3

View all

Shelves

4

View all

Cameras

5

View all

Visitors Tod

128

View Details

System Overview

Cameras Online

4

Cameras Offline

1

Active Visitors

128

Today's Sessions

12

Quick Actions

Add Store

Add Shelf

Connect Camera

View

Recent Activity

Recent Activity

Store "Main Branch" created

10:30 AM

Shelf "Snacks" added in Store 1

09:45 AM

Camera 02 disconnected

09:20 AM

Camera 03 connected

09:10 AM

Shopper Behavior

Track ID	Segment	Dwell	Path	Shelves
1	Comparison Shopper	21.791273	281.34842559454813	2

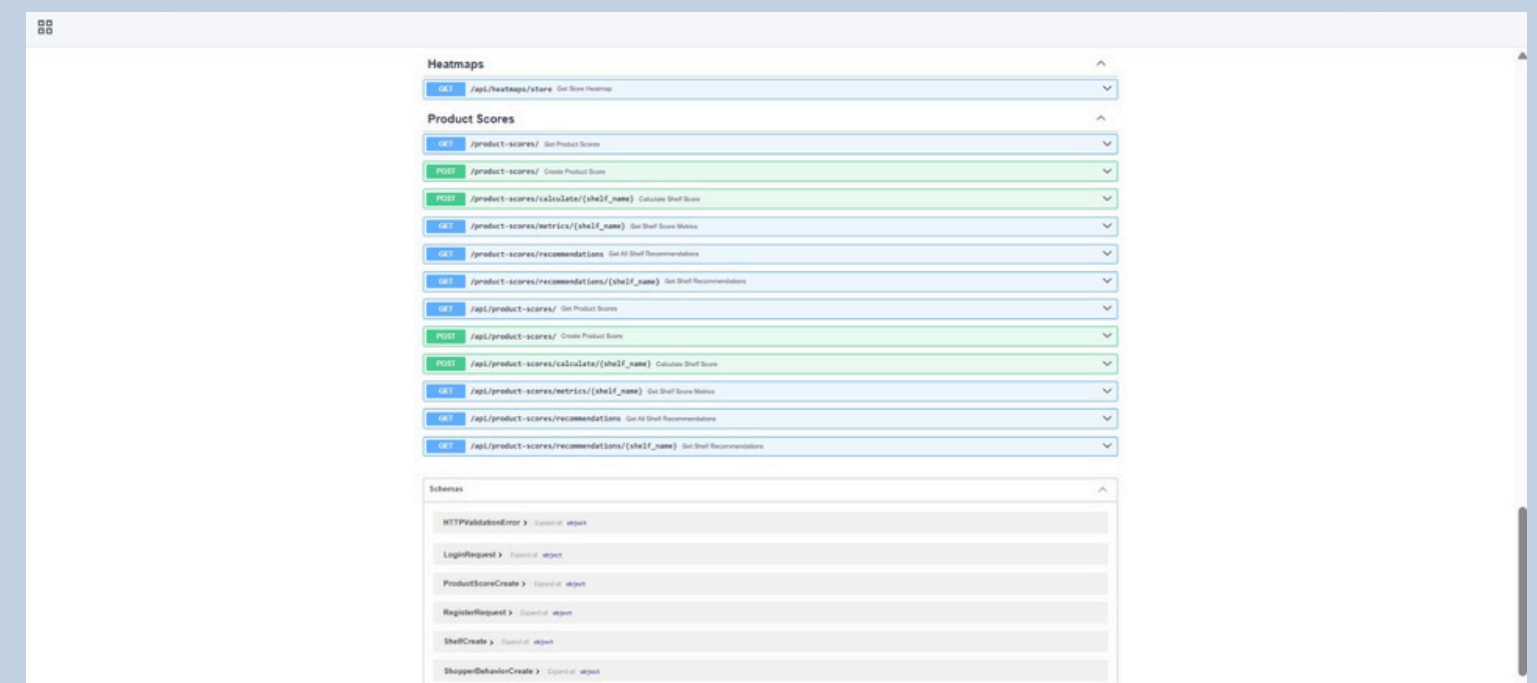
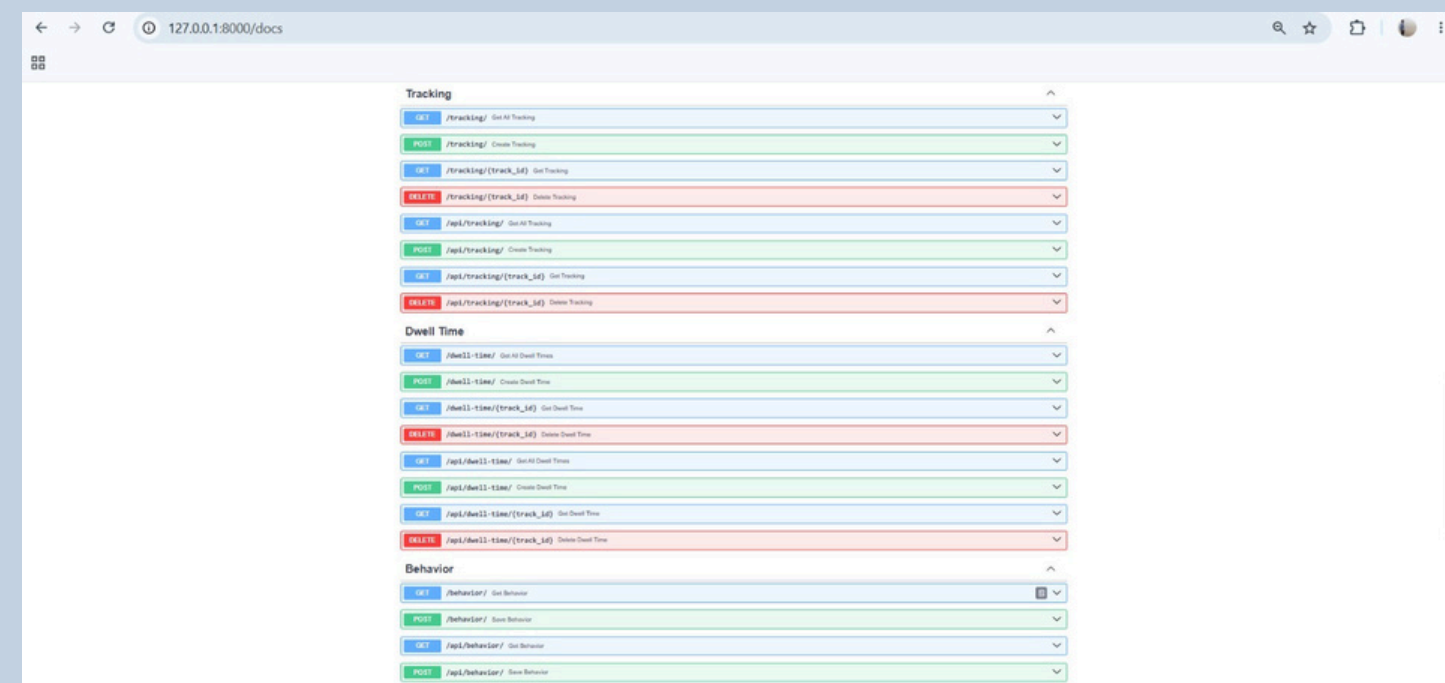
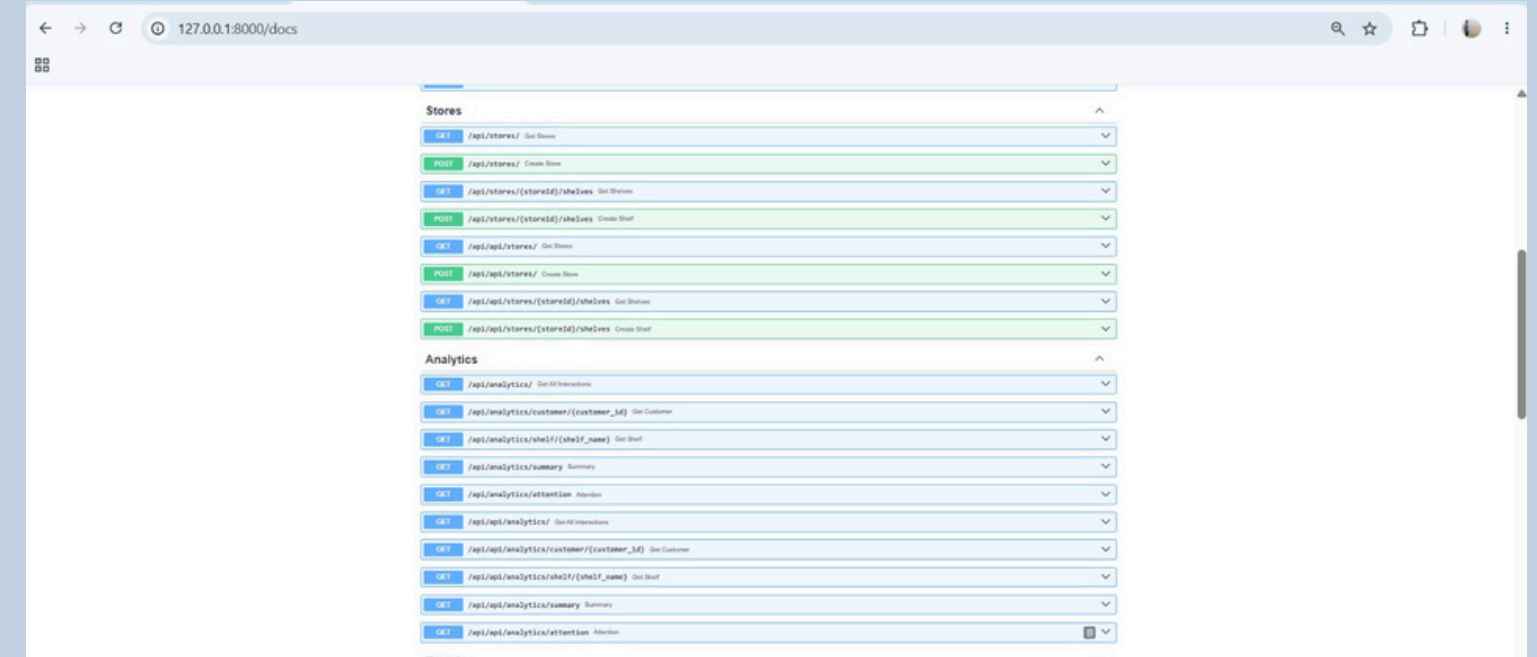
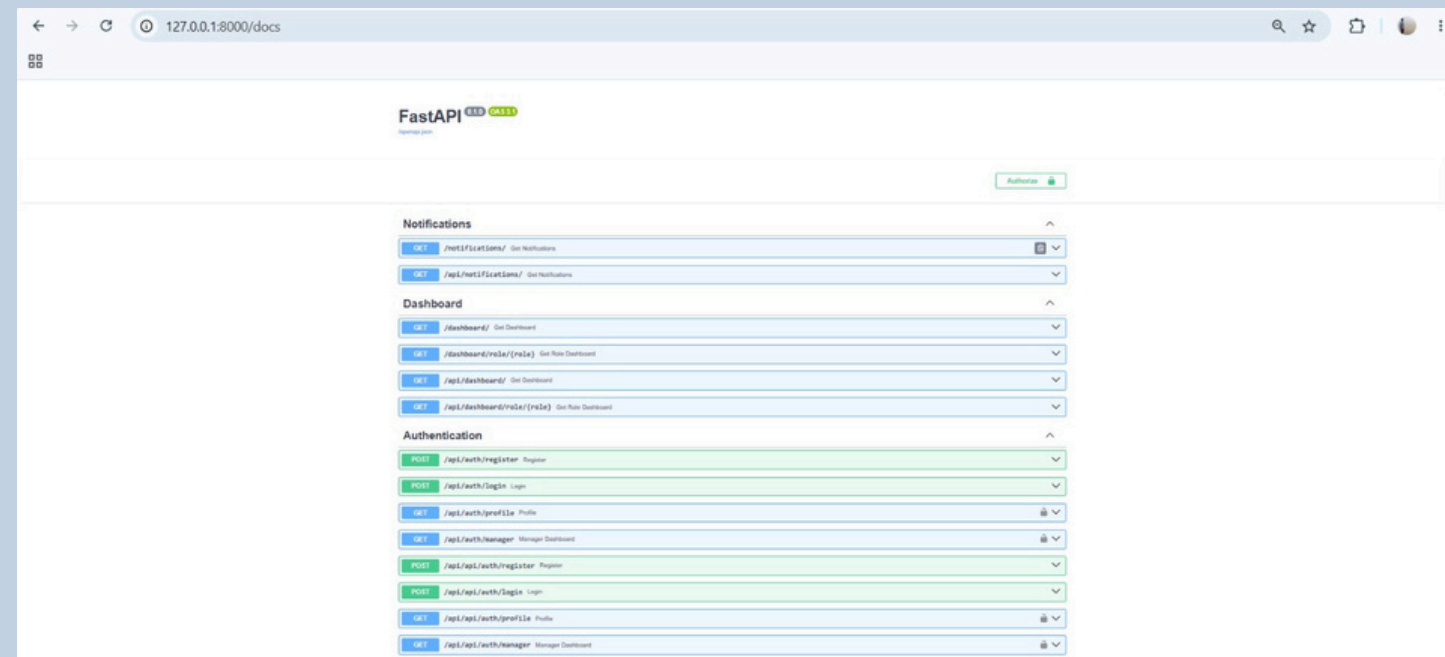
Store Heatmap

Store Heatmap

Today's Shelf Attention Report

Viewing Duration (seconds)

Backend Outputs



Future Enhancements

AI Recommendations: Personalized promotions & smart reminders

Predictive Analytics: Forecast conversions & turnout

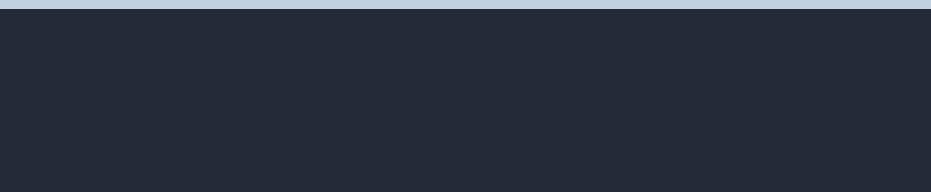
Advanced Visualization: Comparative analysis across stores/events

Expansion: Shopping malls, FMCG, supermarkets



CONCLUSION

Consumer Attention Mapping System enables retailers to optimize shelf placement, improve product visibility, and enhance customer experience.



The background features decorative geometric patterns in the corners. The top-left and bottom-left corners have dark blue and light blue chevron shapes. The top-right and bottom-right corners have more complex patterns with multiple parallel lines in dark blue and light blue, creating a sense of depth and movement.

***THANK
YOU***